

USER MANUAL

HWT901B(TTL)

Robust Inclinometer



Tutorial Link

[Google Drive](#)

Link to instructions DEMO:

[WITMOTION Youtube Channel](#)

[HWT901B Playlist](#)

If you have technical problems or cannot find the information that you need in the provided documents, please contact our support team. Our engineering team is committed to providing the required support necessary to ensure that you are successful with the operation of our AHRS sensors.

Contact

[Technical Support Contact Info](#)

Application

- AGV Truck
- Platform Stability
- Auto Safety System
- 3D Virtual Reality
- Industrial Control
- Robot
- Car Navigation
- UAV
- Truck-mounted Satellite Antenna Equipment

Contents

Tutorial Link	- 2 -
Contact	- 2 -
Application	- 2 -
Contents	- 3 -
1 Introduction	- 4 -
1.1 Warning Statement	- 5 -
2 Use Instructions.....	- 6 -
3 Software Introduction	- 7 -
4 MCU Connection.....	- 8 -

1 Introduction

The HWT901B is a multi-sensor device detecting acceleration, angular velocity, angle as well as magnetic field. The robust housing and the small outline makes it perfectly suitable for industrial retrofit applications such as condition monitoring and predictive maintenance. Configuring the device enables the customer to address a broad variety of use cases by interpreting the sensor data by smart algorithms.

HWT901B's scientific name is AHRS IMU sensor. A sensor measures 3-axis angle, angular velocity, acceleration, magnetic field. Its strength lies in the algorithm which can calculate three-axis angle accurately.

HWT901B is employed where the highest measurement accuracy is required. It offers several advantages over competing sensor:

- Heated for best data availability: new WITMOTION patented zero-bias automatic detection calibration algorithm outperforms traditional accelerometer sensor
- High precision Roll Pitch Yaw (X Y Z axis) Acceleration + Angular Velocity + Angle + Magnetic Field output
- Low cost of ownership: remote diagnostics and lifetime technical support by WITMOTION service team
- Developed tutorial: providing manual, datasheet, Demo video, free software for Windows computer, APP for Android smartphones , and sample code for MCU integration including 51 serial, STM32, Arduino, Matlab, Raspberry Pi, communication protocol for project development
- WITMOTION sensors have been praised by thousands of engineers as a recommended attitude measurement solution

1.1 Warning Statement

- Putting more than 5 Volt across the sensor wiring of the main power supply can lead to permanent damage to the sensor.
- VCC cannot connect with GND directly, otherwise it will lead to the burning of the circuit board.
- For proper instrument grounding: use WITMOTION with its original factory-made cable or accessories.
- Do not access the I2C interface.
- For secondary developing project or integration: use WITMOTION with its compiled sample code.

2 Use Instructions

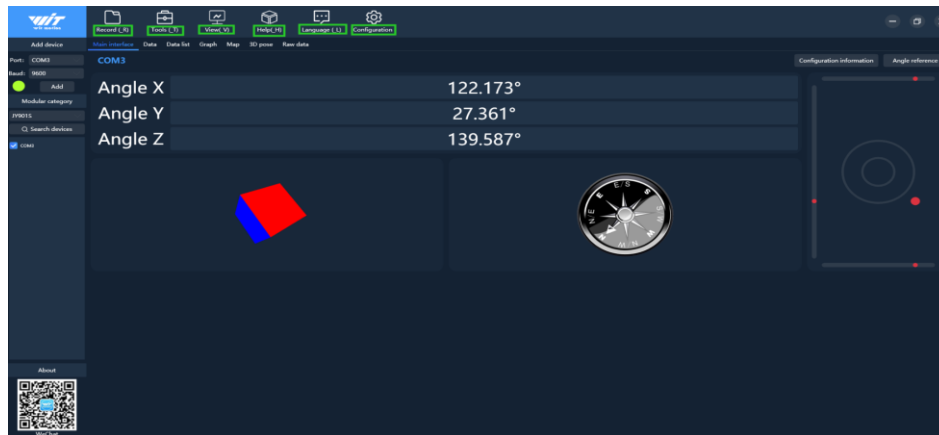
Hit the hyperlink direct to the document or download center:

- [Software and driver download](#)
- [Quick-guide Manual](#)
- [Teaching Video](#)
- [Common Software with detailed instructions](#)
- [SDK\(sample code\)](#)
- [SDK Tutorial Documentation](#)
- [Communication Protocol](#)

3 Software Introduction

[Software function introduction](#)

(Ps. You can check the functions of the software menu from the link.)



SOFTWARE INSTRUCTIONS	1
How to download software and driver	1
PC Software Download:	1
Install Driver	1
Contents	2
1 Use Instructions with PC	4
1.1 Connect with wiring sensor	4
1.2 Connecting with Bluetooth Sensor	11
1.2.1 Select Sensor Model	11
1.2.2 Search Device	11
2 Main Menu	13
2.1 Function configuration	13
2.1.1 Record	13
2.1.2 Data playback	15
2.1.3 Tools	15
2.1.4 View	15
2.1.5 Help	15
2.1.6 Language	15
2.1.7 Configuration	15
2.2 Data Review	17
2.2.1 Main Interface	17
2.2.2 Curve Display	17
2.2.3 Map Function	18
2.2.4 3D Demo	19
2.2.5 Raw data	20
3. Configuration	21
3.1 Read sensor configuration	21
3.2 System settings	22
3.2.1 Reset	22
3.2.2 Sleep and disable sleep setting	23
3.2.3 Alarm setting	24

4 MCU Connection

